

Diabetes Risk Prediction Using Artificial Neural Networks: Baseline Comparison, Hyperparameter Optimization, and Error Analysis

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Abstract— Diabetes is a major public health concern associated with clinical, behavioral, and socioeconomic risk factors. This study used the CDC Diabetes Health Indicators dataset to evaluate artificial neural network models for diabetes risk prediction. Because diabetes-positive cases represented a minority of the dataset, model evaluation emphasized PR-AUC, recall, precision, F1-score, false negative rate, ROC AUC, and Brier score rather than accuracy alone. Baseline models included a majority-class classifier and Logistic Regression. An initial ANN model was trained using one hidden layer with 32 neurons and ReLU activation, followed by hyperparameter optimization across network architecture, activation function, and learning rate. The optimized ANN used one hidden layer with 32 neurons, sigmoid activation, and a learning rate of 0.001, selected based on validation PR-AUC. On the held-out test set, the optimized ANN achieved modest improvement over the initial ANN and performed competitively with Logistic Regression, but the performance gain from additional tuning was limited. Error analysis showed that false negatives remained clinically important, while the recall-oriented threshold increased false positives. These findings suggest that ANN models can support diabetes risk prediction, but simpler models remain competitive for structured tabular health data.

Keywords— Diabetes risk prediction, Artificial neural network, Hyperparameter optimization, Baseline model comparison, Error analysis

I. INTRODUCTION

Diabetes mellitus has become one of the most serious public health concerns in recent decades. According to the Centers for Disease Control and Prevention, more than 40 million Americans are living with diabetes, and type 2 diabetes accounts for approximately 90 to 95 percent of diagnosed cases [1]. Diabetes is strongly associated with modifiable lifestyle and clinical risk factors, including obesity, diet, physical inactivity, hypertension, and cholesterol status. Long-term complications of uncontrolled diabetes include end-stage renal disease, cardiovascular disease, neuropathy, and other microvascular and macrovascular complications, creating substantial burden for patients and healthcare systems [2].

Early identification of individuals at elevated diabetes risk is clinically important because lifestyle modification, screening, and timely intervention may delay disease progression and reduce long-term complications. Machine learning has become increasingly useful for diabetes risk prediction because it can integrate demographic, behavioral, clinical, and self-reported health indicators within a single predictive framework. Prior work using Behavioral Risk Factor Surveillance System data has shown that survey-based machine learning models can provide useful tools for initial type 2 diabetes screening and early intervention [3]. However, in imbalanced healthcare datasets, overall accuracy or discrimination alone may be insufficient, since models can appear strong overall while

still missing a substantial portion of diabetes-positive individuals.

This study builds upon our previous diabetes risk prediction study by introducing Artificial Neural Networks (ANNs) as an alternative modeling approach for identifying individuals at elevated risk of diabetes. Neural networks have been increasingly applied in healthcare prediction because they are capable of learning complex nonlinear relationships among demographic, behavioral, clinical, and socioeconomic risk factors without requiring extensive manual feature engineering [4]. However, previous studies have shown that for structured tabular healthcare datasets, neural networks do not always outperform simpler and more interpretable models such as Logistic Regression by a meaningful margin [5]. Therefore, it remains important to evaluate whether the additional complexity of neural networks is justified for diabetes risk prediction using survey-based health indicators.

This study is guided by three clinical questions. First, can an ANN achieve better predictive performance than baseline approaches, including a Majority Class classifier and Logistic Regression, in identifying individuals with diabetes? Second, does hyperparameter optimization, including modifications to network architecture, activation functions, and learning rates, lead to meaningful improvements over an initial ANN model? Third, how does the optimized ANN compare with the best-performing models from our previous study, and does the increased complexity of a neural network provide sufficient benefit to

justify its use in a healthcare screening setting?

These questions are particularly important because diabetes-positive individuals account for only approximately 13.93% of the observations in the dataset. Under this level of class imbalance, a model may achieve reasonable overall accuracy while still failing to identify a substantial proportion of diabetic individuals. Therefore, this study places greater emphasis on evaluation metrics that are more informative for minority-class detection, including Precision-Recall Area Under the Curve (PR-AUC), Recall, Precision, F1-score, and Brier Score. By focusing on these metrics, we aim to determine whether ANN-based models can improve diabetes risk prediction while maintaining clinically meaningful screening performance.

II. MATERIALS AND METHODS

a. Data Set

This study used the CDC Diabetes Health Indicators dataset obtained from the UCI Machine Learning Repository [6]. The dataset consists of cross-sectional survey data from U.S. adults and includes demographic, behavioral, clinical, and self-reported health variables relevant to diabetes risk. Key variables include body mass index (BMI), blood pressure status, cholesterol status, smoking behavior, physical activity, general health, mental health, physical health, difficulty walking, and socioeconomic indicators such as income and education. The primary outcome variable was *Diabetes_binary*, which indicates whether an individual had diabetes or prediabetes. All remaining health indicator variables were used as predictors for diabetes risk prediction. Most predictors were represented as binary or ordinal variables, making the dataset suitable for supervised classification modeling using artificial neural networks.

b. Study Population

The study population included individuals with complete information on the diabetes outcome and predictor variables. Observations with missing values in key variables were excluded to maintain analytical consistency across model training, validation, and testing. The final analytic sample consisted of 253,680 observations. Diabetes-positive individuals represented 13.93% of the dataset, while diabetes-negative individuals represented 86.07%. This class imbalance motivated the use of evaluation metrics that emphasize minority-class detection rather than overall accuracy alone.

c. Data Preparation

Before model training, the dataset was divided into training, validation, and test sets using stratified sampling. Stratification was used to preserve the proportion of diabetes-positive and diabetes-negative cases across all subsets. The final split consisted of 152,208 observations for training, 50,736 observations for validation, and 50,736 observations for testing. The training set was used for model fitting, the validation set was used for hyperparameter selection and early stopping, and the test set was reserved for final model evaluation. Because neural network models

TABLE 1: DATASET SPLIT AND CLASS DISTRIBUTION FOR NEURAL NETWORK MODELING.

Split	Sample Size	Diabetes = 1 (%)	Diabetes = 0 (%)
Training	152,208	13.93	86.07
Validation	50,736	13.93	86.07
Test	50,736	13.93	86.07

are sensitive to the scale of input variables, all predictor variables were standardized prior to ANN training. Standardization was performed by subtracting the training-set mean and dividing by the training-set standard deviation. The scaler was fit only on the training set and then applied to the validation and test sets to prevent data leakage. Because diabetes-positive cases represented the minority class, model evaluation did not rely on accuracy alone. Instead, performance was assessed using metrics more appropriate for imbalanced classification and healthcare screening, including PR-AUC, recall, precision, F1-score, false negative rate, ROC AUC, and Brier score.

d. Baseline Models

Before developing the Artificial Neural Network (ANN), two baseline models were established to provide reference points for evaluating model performance: a Majority Class classifier and a Logistic Regression model.

The Majority Class classifier represents the simplest possible baseline by predicting every observation as the non-diabetic class. Because the dataset is highly imbalanced, this baseline illustrates why accuracy alone can be misleading when evaluating diabetes prediction models. Although such a classifier may achieve relatively high accuracy, it cannot effectively identify diabetes-positive individuals and therefore has limited practical value in a healthcare screening setting.

Logistic Regression was selected as the primary supervised baseline model because it is widely used in healthcare prediction tasks due to its simplicity, interpretability, and strong performance on structured tabular data. Following the approach used in Assignment 1, the model was trained using standardized predictor variables and balanced class weights to address class imbalance. Logistic Regression served as the main benchmark against which the ANN models were compared throughout this study.

e. Initial ANN Model

The initial Artificial Neural Network (ANN) was developed as a simple feed-forward neural network to establish a baseline neural network architecture for diabetes risk prediction. The model used all standardized predictor variables as inputs and consisted of a single hidden layer with 32 neurons using the ReLU activation function, followed by a one-neuron output layer with a sigmoid activation function to generate the predicted probability of diabetes.

The model was implemented using TensorFlow and Keras. Binary cross-entropy was selected as the loss function because diabetes prediction is a binary classification task, and the Adam optimizer was used to

update model weights during training. To improve generalization and reduce overfitting, early stopping was applied based on validation loss with a patience of five epochs, allowing the best-performing model weights to be restored automatically. Training was performed using a batch size of 32 and a maximum of 20 epochs.

Because the dataset was highly imbalanced, threshold selection was considered an important component of model development. Preliminary evaluation indicated that the default classification threshold of 0.50 resulted in relatively low sensitivity for identifying diabetes-positive individuals. Therefore, several threshold values were evaluated, and a threshold of 0.15 was selected to improve recall while maintaining a reasonable balance between precision and false positives. This threshold was chosen to better align the model with the screening-oriented objective of identifying individuals at elevated diabetes risk.

The initial ANN served as the baseline neural network architecture for subsequent hyperparameter tuning and model optimization.

f. Hyperparameter Optimization

After training the initial ANN model, hyperparameter optimization was performed to evaluate whether alternative neural network architectures could improve model performance. Because the dataset was imbalanced and diabetes-positive cases represented the minority class, validation PR-AUC was used as the primary criterion for selecting the optimized model. PR-AUC was chosen because it better reflects the model's ability to identify positive cases under class imbalance than accuracy alone.

A targeted hyperparameter search was conducted over several candidate ANN configurations. The search varied three main components: hidden layer architecture, activation function, and learning rate. The tested hidden layer architectures included one-hidden-layer models with 32 or 64 neurons and two-hidden-layer models with 32–16 or 64–32 neurons. These architectures were selected to compare simpler networks with moderately more complex models while avoiding overly large architectures for a low-dimensional tabular dataset. The activation functions included ReLU, tanh, and sigmoid, and the learning rates included 0.001 and 0.0001.

For each candidate model, the network was trained using the Adam optimizer and binary cross-entropy loss. Early stopping was applied based on validation loss to reduce overfitting and unnecessary training. Each model was evaluated on the validation set using accuracy, ROC AUC, PR-AUC, recall, precision, F1-score, false negative rate, and Brier score. The model with the highest validation PR-AUC was selected as the optimized ANN model.

After the optimized architecture was selected using the validation set, final performance was evaluated on the held-out test set. The test set was not used during hyperparameter selection, ensuring that the final evaluation reflected out-of-sample model performance.

g. Optimized ANN Model

Based on the hyperparameter search results, the optimized ANN model was selected using validation PR-AUC as the primary criterion. The best-performing configuration consisted of one hidden layer with 32 neurons, sigmoid activation, and a learning rate of 0.001. This model achieved the highest validation PR-AUC among the candidate configurations, indicating the best precision-recall trade-off for identifying diabetes-positive cases in the validation set.

Although deeper architectures were evaluated, they did not substantially improve validation PR-AUC compared with the simpler one-hidden-layer model. Therefore, the final optimized ANN favored a simpler architecture that achieved strong validation performance while reducing unnecessary model complexity.

The optimized model used a sigmoid output layer to produce predicted probabilities for the binary diabetes outcome. Binary cross-entropy was used as the loss function, and the Adam optimizer was used for model training. Early stopping was applied based on validation loss to reduce overfitting and restore the best-performing model weights.

After selecting the optimized architecture using the validation set, the model was evaluated on the held-out test set. The test set was not used during hyperparameter optimization and was reserved for final performance assessment. A decision threshold of 0.15 was used to convert predicted probabilities into binary classifications, consistent with the screening-oriented goal of prioritizing recall and reducing false negatives for diabetes-positive individuals.

III. MODELING RESULTS

a. Baseline Model Performance

Baseline models were evaluated to provide reference points for interpreting the neural network results. The majority-class baseline assigned all observations to the non-diabetic class. As shown in Table 2, this baseline achieved relatively high accuracy (0.861) because the dataset was imbalanced and most observations were non-diabetic. However, it failed to identify any diabetes-positive cases, resulting in a recall of 0.000 and a false negative rate of 1.000. This demonstrates that accuracy alone is not sufficient for evaluating model performance in this screening-oriented task.

Logistic Regression was used as a simple supervised baseline model. Compared with the majority-class baseline, Logistic Regression provided meaningful discrimination between diabetes-positive and diabetes-negative individuals, achieving a ROC AUC of 0.820 and PR-AUC of 0.393. The model also achieved a recall of 0.760, indicating that it identified a substantial proportion of diabetes-positive individuals. However, precision remained relatively low at 0.311, reflecting the expected trade-off between sensitivity and false positives in an imbalanced classification setting.

TABLE 2: BASELINE MODEL PERFORMANCE ON THE TEST SET.

Model	Acc.	ROC-AUC	PR-AUC	Recall	Precision	F1	FNR
Majority Class	0.861	0.500	0.139	0.000	0.000	0.000	1.000
Logistic Regression	0.732	0.820	0.393	0.760	0.311	0.441	0.240

b. Initial ANN Model Performance

The initial ANN model used a simple architecture with one hidden layer containing 32 neurons, followed by a sigmoid output layer for binary classification. The model was evaluated on the held-out test set using a threshold of 0.15, consistent with the screening-oriented goal of prioritizing recall for diabetes-positive cases.

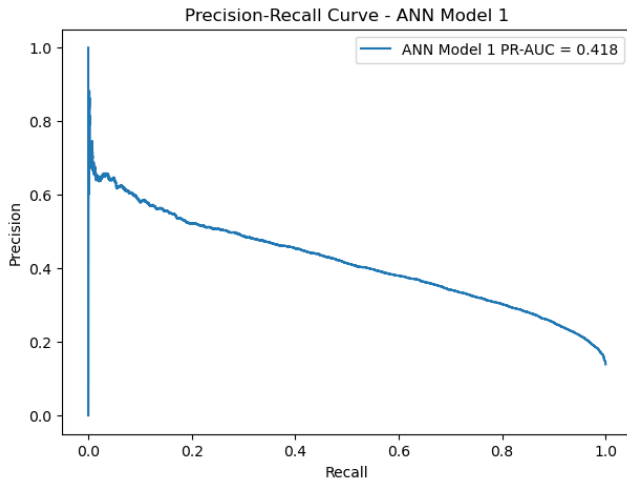


Fig. 1: Precision-Recall Curve for the Initial ANN Model

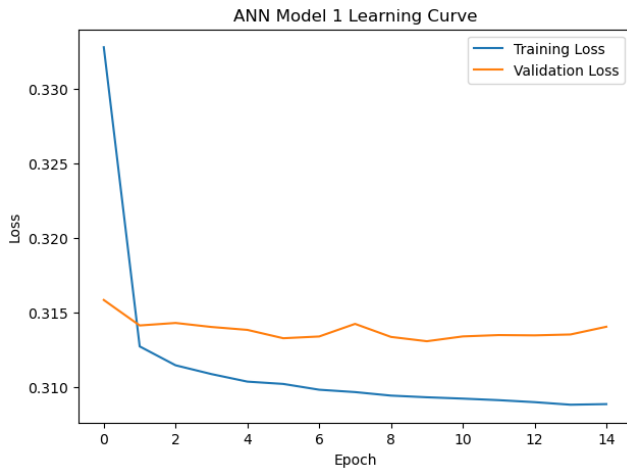


Fig. 2: Training and Validation Loss Across Epochs for the Initial ANN Model

As shown in Table 3, the initial ANN achieved a ROC AUC of 0.825 and PR-AUC of 0.417. At the selected threshold, the model achieved a recall of 0.777, precision of 0.313, and F1-score of 0.446. These results indicate that even a relatively simple neural network was able to learn meaningful patterns from the health indicator variables and substantially outperformed the majority-class baseline in identifying diabetes-positive individuals.

TABLE 3: INITIAL ANN MODEL PERFORMANCE ON THE TEST SET.

Model	Threshold	Acc.	ROC-AUC	PR-AUC	Recall	Precision	F1	Brier
Initial ANN	0.15	0.731	0.825	0.417	0.777	0.313	0.446	0.098

c. Hyperparameter Optimization Results

Hyperparameter optimization was conducted to determine whether alternative ANN configurations could improve validation performance. Candidate models varied by hidden layer architecture, activation function, and learning rate. Because diabetes-positive cases represented the minority class, validation PR-AUC was used as the primary selection criterion.

Among the tested configurations, the best validation performance was achieved by a one-hidden-layer ANN with 32 neurons, sigmoid activation, and a learning rate of 0.001. This configuration achieved a validation PR-AUC of 0.434, ROC AUC of 0.828, recall of 0.768, precision of 0.315, and F1-score of 0.446. Although several candidate models produced similar validation performance, this model was selected as the optimized ANN because it achieved the highest validation PR-AUC.

The hyperparameter search results also suggest that increasing network complexity did not substantially improve performance. Models with deeper architectures or alternative activation functions produced very similar PR-AUC values, indicating that a simpler ANN architecture was sufficient for this structured tabular dataset.

TABLE 4: VALIDATION PERFORMANCE OF ONE-HIDDEN-LAYER ANN MODELS WITH 32 NEURONS UNDER DIFFERENT ACTIVATION FUNCTIONS AND LEARNING RATES.

Architecture	Activation	LR	Epochs	PR-AUC	ROC-AUC	Recall	Precision	F1
(32)	ReLU	0.0010	14	0.429	0.827	0.768	0.314	0.445
(32)	ReLU	0.0001	50	0.432	0.828	0.770	0.317	0.449
(32)	Tanh	0.0010	14	0.429	0.827	0.763	0.319	0.450
(32)	Tanh	0.0001	50	0.428	0.827	0.758	0.317	0.447
(32)	Sigmoid	0.0010	24	0.434	0.828	0.768	0.315	0.446
(32)	Sigmoid	0.0001	50	0.431	0.828	0.762	0.318	0.449

d. Optimized ANN Model Performance

Based on validation PR-AUC, the optimized ANN model was selected as the one-hidden-layer model with 32 neurons, sigmoid activation, and a learning rate of 0.001. After selecting this configuration using the validation set, the optimized model was evaluated on the held-out test set. The test set was not used during hyperparameter optimization.

As shown in Table 5, the optimized ANN achieved a test ROC AUC of 0.826 and PR-AUC of 0.419. At the threshold of 0.15, the model achieved a recall of 0.774, precision of 0.316, and F1-score of 0.449. Compared with the initial ANN model, the optimized ANN showed a small improvement in accuracy, PR-AUC, precision, and F1-score, while recall decreased slightly from 0.777 to 0.774. Overall, the optimized ANN provided a modest improvement over the initial ANN, suggesting that hyperparameter tuning improved the precision-recall balance but did not dramatically change overall performance.

Compared with the Logistic Regression baseline, the optimized ANN achieved slightly higher ROC AUC, PR-AUC, precision, and F1-score, while maintaining similar recall. These results suggest that the optimized ANN performed competitively with the simple supervised baseline, although the magnitude of improvement was modest.

TABLE 5: OPTIMIZED ANN MODEL PERFORMANCE ON THE HELD-OUT TEST SET.

Model	Threshold	Acc.	ROC-AUC	PR-AUC	Recall	Precision	F1	Brier
Optimized ANN	0.15	0.736	0.826	0.419	0.774	0.316	0.449	0.098

e. Model Comparison

Table 6 summarizes the final comparison among the initial ANN, the optimized ANN, and the best-performing model from Phase I, XGBoost. Overall, hyperparameter tuning produced a modest improvement over the initial ANN rather than a dramatic change in performance. Compared with the initial ANN, the optimized ANN achieved slightly higher ROC AUC and PR-AUC, indicating somewhat stronger discrimination and minority-class ranking performance.

The main practical difference between the two ANN models was in their overall screening profile. At the retained threshold of 0.15, recall changed only slightly from 0.777 in the initial ANN to 0.774 in the optimized ANN, while precision improved marginally from 0.313 to 0.316. F1-score remained very similar between the two models. This pattern suggests that hyperparameter optimization did not radically change the model, but it did produce a modest and consistent improvement in discrimination, precision-recall performance, and probability calibration.

The optimized ANN also compares favorably with the best-performing model from Phase I. As shown in Table 6, the optimized ANN achieved a ROC AUC of 0.826 and a PR AUC of 0.419, both of which were very close to the XGBoost results from the previous phase. Although recall was slightly lower than that of XGBoost at the same threshold, the optimized ANN maintained comparable precision, F1-score, and Brier score. These results suggest that the optimized ANN performed competitively with the strongest model from the earlier phase and that neural network optimization provided modest gains over the initial ANN without sacrificing overall stability.

Figure 3 reinforces the ANN-to-ANN comparison. The precision-recall curves of the initial and optimized ANN models remain close across most of the recall range, showing that the optimized ANN is not a radically different classifier. However, the optimized model stays slightly above the initial model over much of the curve and therefore achieves a higher PR AUC overall. In the context of imbalanced diabetes prediction, this improvement is meaningful because PR AUC more directly reflects the model's ability to prioritize diabetes-positive cases than accuracy alone.

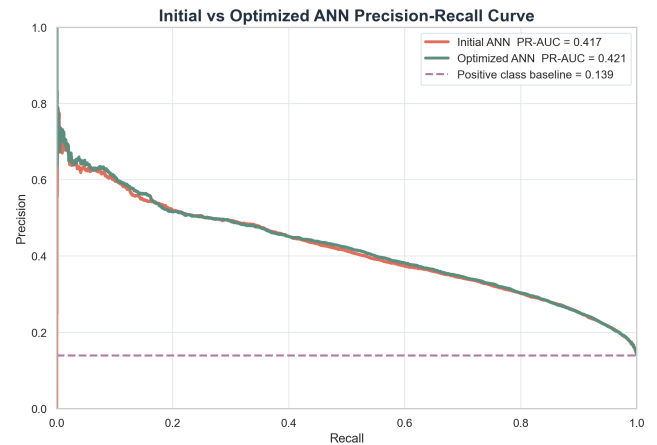


Fig. 3: Precision-recall curves for the initial and optimized ANN models. The optimized ANN achieved a slightly higher PR-AUC, indicating modest improvement in minority-class ranking performance.

f. Error Analysis

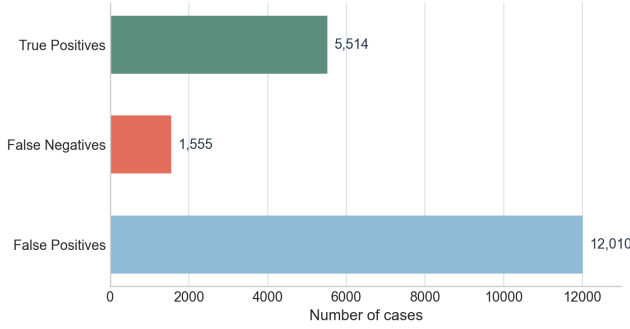
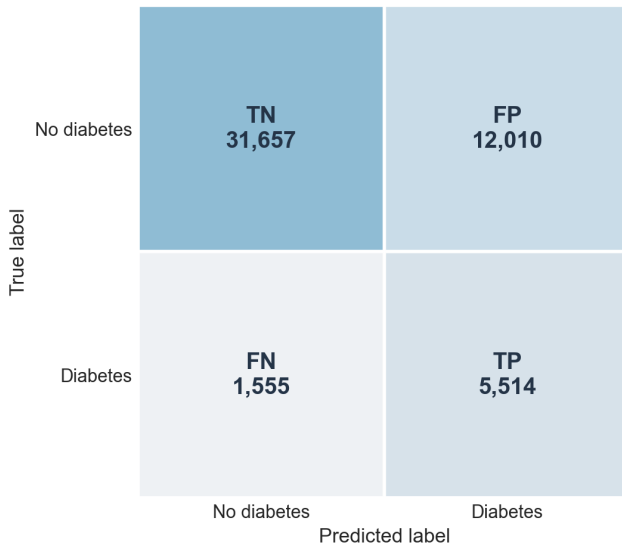
To better understand the behavior of the final model, we examined classification errors separately from overall performance metrics. The optimized ANN was evaluated at a threshold of 0.15, which was retained as a recall-oriented cutoff rather than an F1-optimal cutoff. Although thresholds around 0.20 can yield slightly higher F1-scores, they also reduce recall. Because the goal of this project is screening rather than high-precision diagnosis, we prioritized identifying more diabetes-positive individuals even at the cost of additional false positives.

The error analysis in Figure 4 shows that the final ANN correctly identified 5,514 diabetes-positive individuals while missing 1,555, corresponding to a false negative rate of 0.220. In other words, approximately 22 percent of diabetes-positive cases in the test set were still missed by the model. This is not negligible, but it reflects a lower false-negative burden than would be expected under a more conservative threshold. In a screening context, these false negatives are the most clinically important errors, because they represent individuals who may not be flagged for further evaluation despite elevated risk.

At the same time, the model produced 12,010 false positives, which is the main trade-off of using a lower threshold. This means that the gain in recall was achieved by widening the net of positive predictions, thereby capturing more true diabetes cases while also labeling more non-diabetic individuals as high risk. From a screening perspective, this trade-off is defensible: false positives may lead to additional follow-up or testing, but false negatives represent missed opportunities for early detection. For this reason, the optimized ANN should be interpreted as a recall-first screening model rather than a high-specificity diagnostic tool. Figure 5 provides the corresponding confusion matrix, showing the full distribution of true negatives, false positives, false negatives, and true positives under the retained threshold of 0.15.

TABLE 6: COMPARISON OF THE INITIAL ANN, OPTIMIZED ANN, AND THE BEST-PERFORMING PHASE I MODEL (XGBOOST).

Model	Architecture / Type	Threshold	ROC AUC	PR AUC	Recall	Precision	F1-score	Brier Score
XGBoost (Phase I)	Tree-based ensemble	0.15	0.829	0.428	0.772	0.318	0.451	0.097
Initial ANN	32	0.15	0.825	0.417	0.777	0.313	0.446	0.098
Optimized ANN	32	0.15	0.826	0.419	0.774	0.316	0.449	0.098

**Fig. 4:** Error breakdown of the optimized ANN at threshold 0.15. False positives remained more common than false negatives, reflecting the recall-oriented screening threshold.**Fig. 5:** Confusion matrix of the optimized ANN at threshold 0.15. The model identified many diabetes-positive cases while still missing a smaller subset of positive cases.

IV. DISCUSSION

The modeling results highlight an important finding that is consistent with prior literature: for structured tabular healthcare data, neural networks do not necessarily outperform simpler models by a meaningful margin [5]. The optimized ANN achieved a ROC-AUC of 0.826 and PR-AUC of 0.419, which were only slightly higher than those of the Logistic Regression baseline (0.820 and 0.393, respectively). This modest improvement suggests that the additional complexity of a neural network may not always be justified when the predictor variables consist primarily of binary and ordinal health indicators.

The hyperparameter search showed that increasing network depth and changing activation functions did not substantially improve performance. The best configuration was a simple one-hidden-layer model with 32 neurons and

sigmoid activation, which performed comparably to or better than deeper architectures in this study. This finding suggests that model complexity was not the primary limiting factor for predictive performance on this dataset. The marginal gains from tuning were observed mainly in PR-AUC and precision rather than recall, indicating that the optimized ANN achieved a slightly better precision-recall balance without fundamentally changing the model's behavior.

A consistent challenge throughout this study was the substantial class imbalance, with diabetes-positive cases representing only 13.93% of the dataset. As demonstrated by the Majority Class classifier, accuracy alone can be highly misleading under such conditions [7]. For the ANN models, a classification threshold of 0.15 was selected to improve recall and better identify diabetes-positive individuals. This decision reflects a deliberate choice to prioritize sensitivity over precision in a population screening context. Although lowering the threshold increased the number of false positives, it also reduced the likelihood of missing individuals at elevated risk of diabetes. In a screening-oriented healthcare setting, false negatives are often considered more clinically consequential than false positives because missed cases may delay further evaluation, intervention, and disease management.

Compared with the best-performing models from our previous study, the optimized ANN achieved similar predictive performance. XGBoost achieved the strongest overall performance in the previous analysis, with a ROC-AUC of 0.829 and PR-AUC of 0.428, while the optimized ANN achieved 0.826 and 0.419, respectively. These differences were relatively small and are unlikely to be clinically meaningful in a diabetes screening setting. Although the ANN achieved competitive performance, it did not clearly outperform the best tree-based model.

One potential advantage of Logistic Regression and XGBoost is interpretability. Logistic Regression provides directly interpretable coefficients, while XGBoost can provide feature importance measures that help identify influential predictors. In contrast, neural networks operate as less transparent models, making it more difficult to understand how individual variables contribute to predictions. Therefore, when predictive performance is similar, model interpretability may become an important consideration for real-world healthcare applications.

Several limitations should be acknowledged. First, the dataset is cross-sectional and therefore does not provide information about temporal relationships or disease progression. Second, the predictor variables are primarily self-reported health indicators and may be subject to reporting bias or measurement error. Third, although neural networks require greater computational resources than simpler models such as Logistic Regression, only modest

performance improvements were observed in this study.

Future work could explore additional neural network architectures and imbalance-aware approaches to improve minority-class detection. External validation using independent datasets would also be necessary before deploying these models in a real-world clinical screening settings. Furthermore, evaluating model performance across demographic and socioeconomic subgroups may provide additional insights into the consistency of ANN-based diabetes risk prediction across different populations.

V. CONCLUSION

This study evaluated artificial neural network models for diabetes risk prediction using the CDC Diabetes Health Indicators dataset. Because diabetes-positive cases represented a minority of the sample, model evaluation emphasized PR-AUC, recall, precision, F1-score, false negative rate, ROC AUC, and Brier score rather than accuracy alone.

The majority-class baseline achieved high accuracy but failed to identify any diabetes-positive cases, demonstrating that accuracy alone is misleading in this imbalanced classification task. Logistic Regression provided a stronger supervised baseline, achieving meaningful discrimination and high recall. The initial ANN model also performed well, showing that a simple neural network architecture could learn useful patterns from demographic, clinical, behavioral, and self-reported health indicators.

Hyperparameter optimization selected a one-hidden-layer ANN with 32 neurons, sigmoid activation, and a learning rate of 0.001 based on validation PR-AUC. On the held-out test set, the optimized ANN achieved slightly better PR-AUC, precision, and F1-score than the initial ANN, while maintaining a similar level of recall. However, the improvement was modest.

Compared with the best-performing model from the previous study, the optimized ANN achieved similar predictive performance to XGBoost, suggesting that neural networks can be a viable alternative for diabetes risk prediction. However, given the modest performance gains over simpler models, the added complexity of a neural network may not always be justified for structured tabular health data. Future work should focus on improving false negative detection and evaluating model performance across demographic subgroups before deploying these models in a real clinical screening context.

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